

```
import numpy as np
import scipy.integrate as spi
import matplotlib.pyplot as plt

from astropy import units as u
```

```
import pandas as pd
```

```
import tensorflow as tf
```

Computing the Cosmos

Bridging Theoretical Physics and
Numerical Methods with Python

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› Python4Physics 2025 | UC Berkeley, USA

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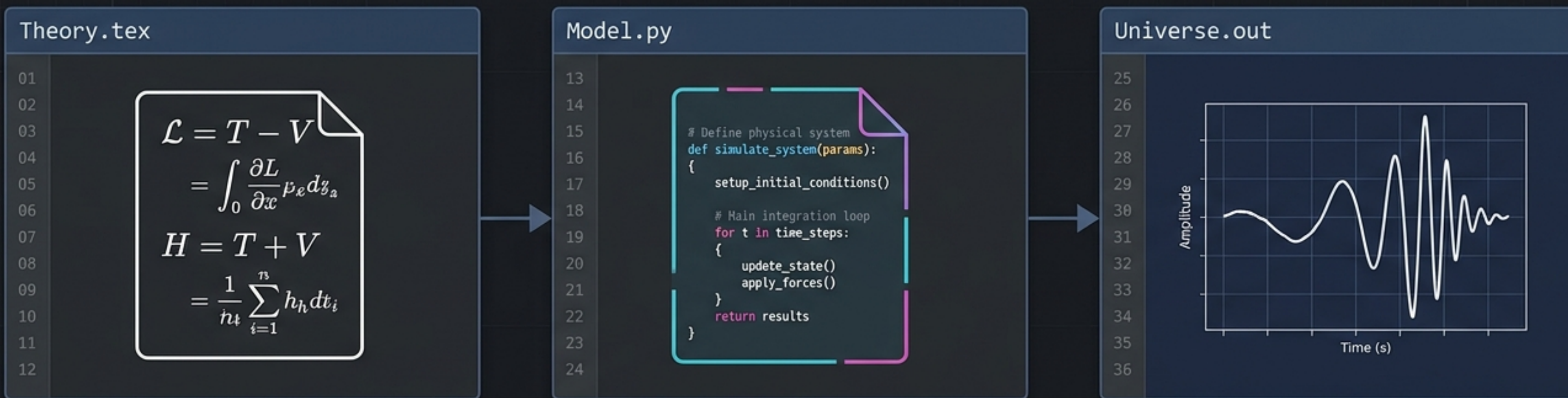
```
import torch
```

```
import h5py
```

```
import tensorflow as tf
```

```
import numba
```

The Pedagogical Philosophy



Computation is not just a tool; it is a unique lens for discovery in physics.

The Mission

Merging the elegance of physical laws with the rigorous power of computational thinking to solve modern physics problems numerically.

The Foundation

Built upon intensive study of Python, NumPy, Matplotlib, and SciPy to rigorize experimental analysis and statistical inference.

The Foundational Toolkit

Mathematical Foundations

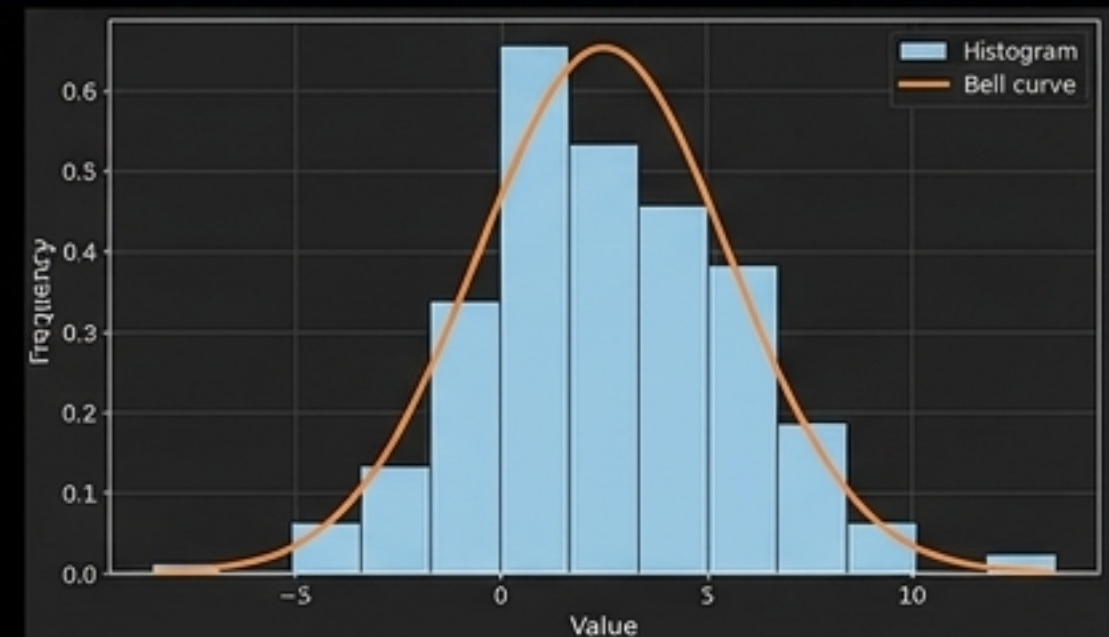
- **Calculus:** Translating continuous systems into numerically solvable differential equations.
- **Statistics:** Drawing on Kenneth McElvain's lectures—utilizing probability distributions, standard deviations, and histograms for rigorous empirical data analysis.

Key Insight: Discretization is the fundamental bridge between theoretical math and applied modeling.

Implementation

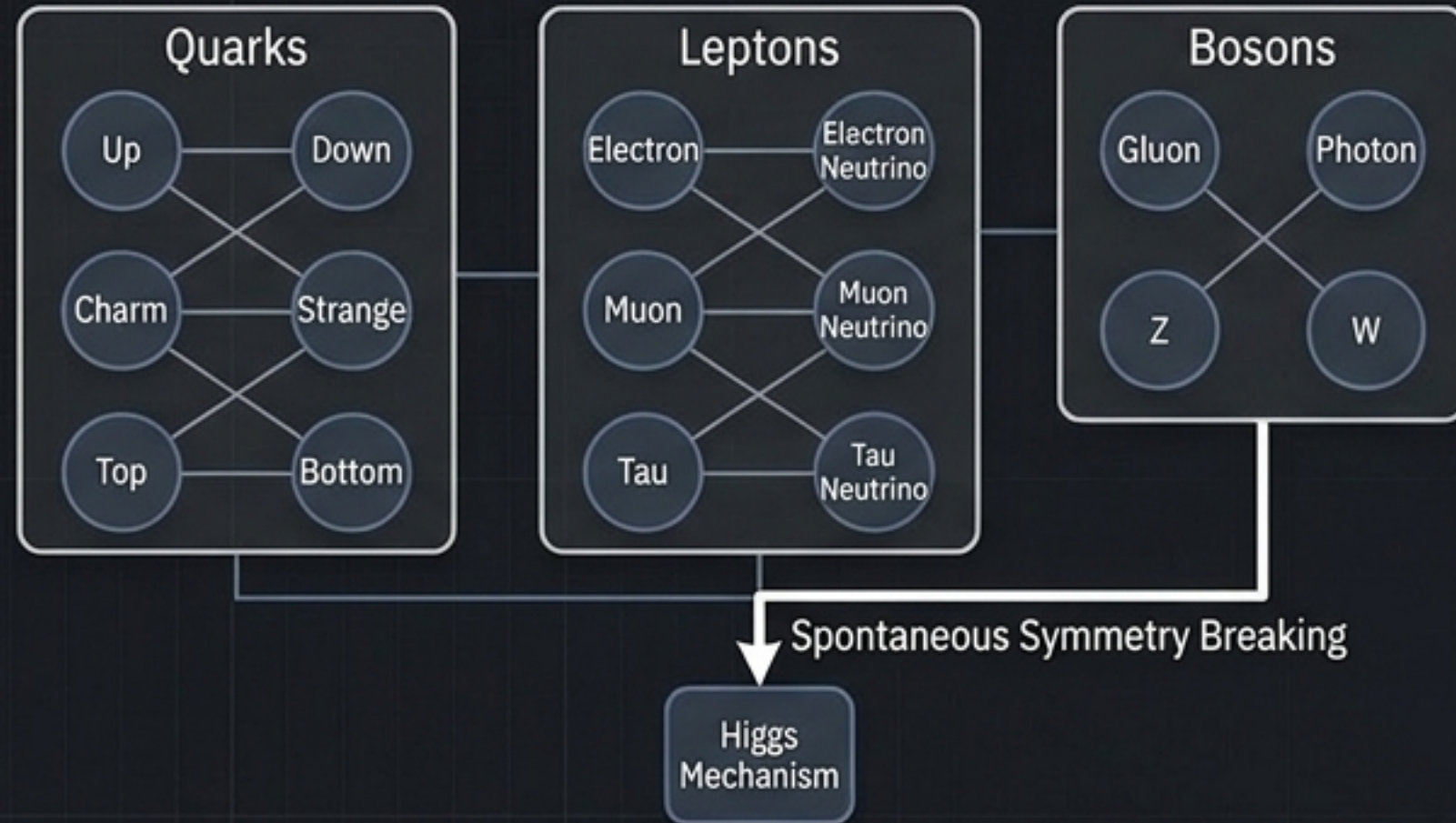
```
1 import matplotlib.pyplot as plt
2 import numpy as np
3
4 pi= np.pi
5 def sqrt(x): return np.sqrt(x)
6 def exp(x): return np.exp(x)
7 def sin(x): return np.sin(x)
8 def cos(x): return np.cos(x)
9 def log(x): return np.log(x)
10 def poly0(x): return (pow(x,3)/6.) + (pow(x,4)/8.0)
11
12 """
13 *****
14 | solving linear differential equations
15 *****
16 """
17
18 def lin_diff_eq(g, f0, xs, a):
```

Output



The Quantum Realm

The Standard Model



Simulation Sandbox

```
// Particle Decay Simulation & Histogram Plotting
11
12 p_B = np.random.normal(loc=0, scale=1, size=1000)
13
14 fig = plt.figure()
15
16 ax = plt.axes([0.15, 0.15, 0.75, 0.75])
17
18 ax.hist(p_B, bins=50, alpha=0.7, density=False);
```

- **The Physical Phenomenon:** Modeling a 2-body neutron decay process.
- **The Computational Method:** Utilizing Monte Carlo methods and NumPy random distributions.
- **The Physics-Code Connection:** The script rigorously applies the conservation of momentum and energy, translating abstract theoretical principles directly into precise numerical constraints.

Statistical Mechanics & Emergent Behavior

Theoretical Framework:

- Magnetism, spin interactions, and symmetry breaking (via Felipe Ortega Gama).

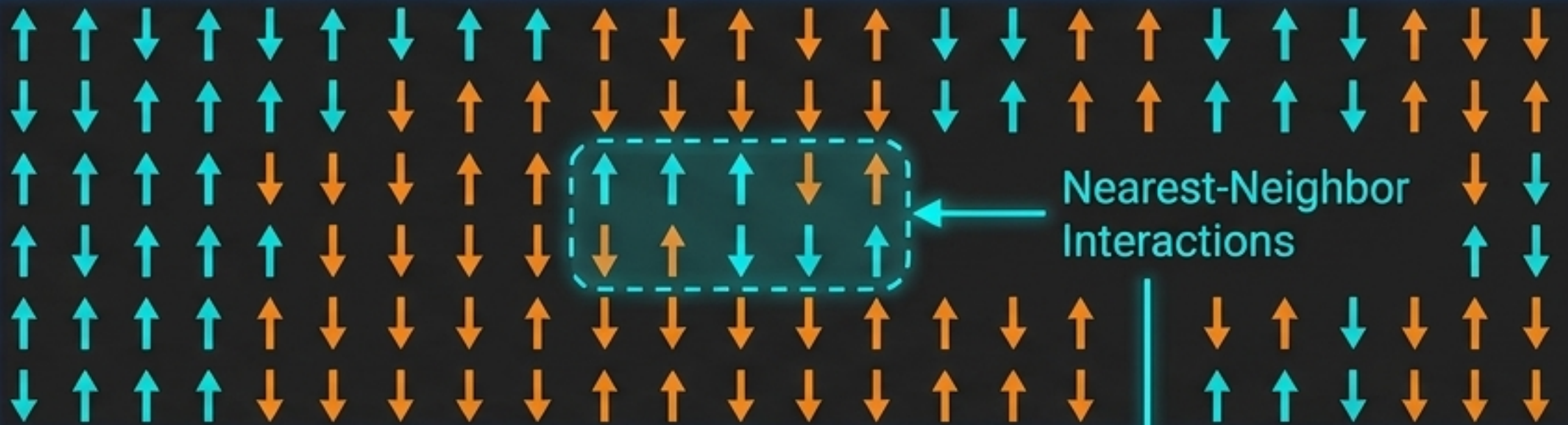
Computational Translation:

- Simulating both nearest- and next-nearest-neighbor couplings through efficient array rotation.

Emergent Insight:

- Highly localized microscopic states yield macroscopic degeneracy and the collective behavior of condensed matter spin systems.

The Physics



The Code

```
11 def rotate_vec(vec0):
12     """
13     this piece of code rotates an array (vec0)
14     by one to the left
15     """
16     N=len(vec0)
17     tmp = np.zeros(N)
18
19     tmp[0:N-1]= vec0[1:N]
20
21     tmp[N-1]= vec0[0]
22
23     return tmp
```

The Macroscopic Universe

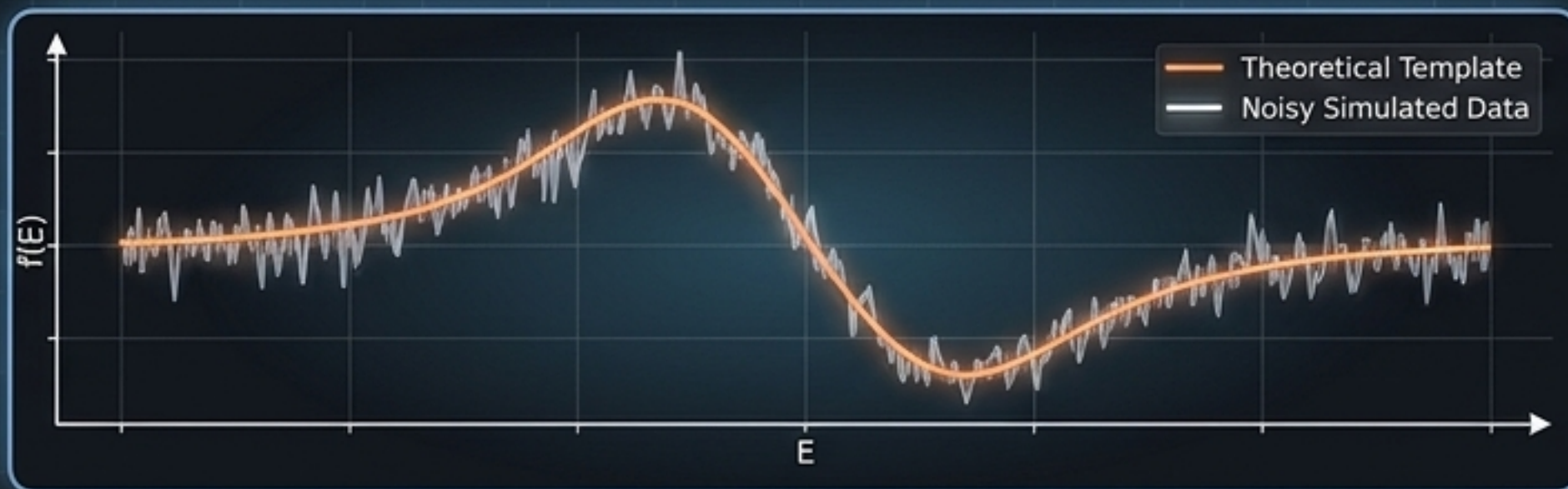
Layer 1: Theory

$$f(E) = \frac{k}{(E - M)^2 + (\text{Gamma} / 2)^2}$$

Layer 2: Code

```
def P_BW(M, Gamma, E):  
    tmp1 = (E - M)**2  
    tmp2 = (Gamma / 2)**2  
    tmp3 = Gamma / (2 * pi)  
    return tmp3 / (tmp1 + tmp2)
```

Layer 3: Result



- **Domain:** Resonances & Gravitational Waves.
- **The Technique:** Leveraging Python's SciPy library for advanced curve fitting.
- **Application:** Mirroring the exact analytical approaches used by real-world LIGO physicists to detect cosmic collisions.

Frontiers of Physics

IDE Terminal

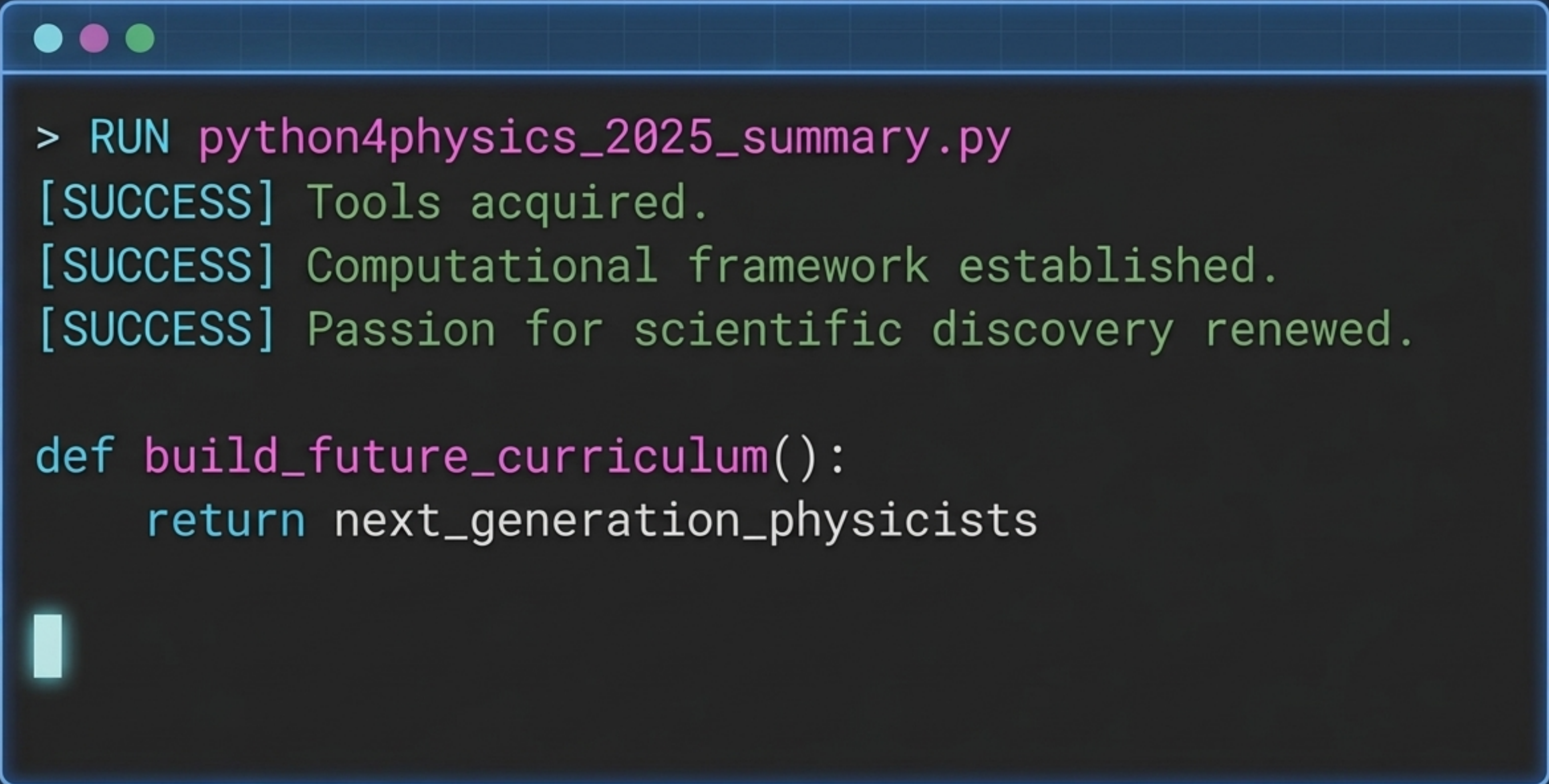
1 The Frontier: Axions and Dark Matter

2
3
4
5 **The Challenge:** Designing computational
6 filters to detect nearly invisible quantum-
6 scale particles using massive cosmic-scale
7 datasets.
8
9

10
11
12 **Synthesis:** Elegantly bridging the cosmos
13 and the quantum realm—the two extremes
14 of modern physics united by computational
15 data analysis.
16
17
18

The Physics-Computation Matrix

Physics Domain	Governing Theory	Computational Method (Python)	Core Insight
Quantum Particle	Standard Model / Conservation Laws	NumPy / Monte Carlo Sim	Predicting decay distributions
Condensed Matter	Ising Model / Magnetism	Array Manipulation (rotate_vec)	Modeling symmetry breaking
Astrophysics	General Relativity / Resonances	SciPy / Curve Fitting	Extracting signals from noise
Cosmology	Dark Matter / CMB	Big Data Statistical Inference	Bridging macro/micro scales



```
> RUN python4physics_2025_summary.py  
[SUCCESS] Tools acquired.  
[SUCCESS] Computational framework established.  
[SUCCESS] Passion for scientific discovery renewed.
```

```
def build_future_curriculum():  
    return next_generation_physicists
```



With gratitude to UC Berkeley, the program mentors, and the Python4Physics community for providing the blueprint to merge theory and computation in higher education.